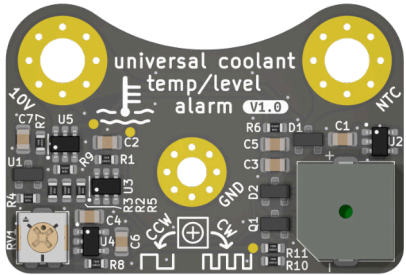


universal coolant temp/level alarm V1.0



github
UCTLA

Assembly instructions

This board is screwed onto the existing cooling temperature gauge on the instrument cluster and beeps as soon as the **temperature** is **too high** or the **coolant level** sensor detects **too little water**.

With "ignition on" the red LED flashes as usual, but no annoying buzzer sounds.

Compatibility

All vehicles with a **triple-screwed temperature gauge** can be equipped with this board.

191 919 511 A ✓
251 919 511 A ✓
321 919 511 K ✓

T3/vanagon 1979 - 1992	Jetta 2 syncro 1986 - 1991
Caddy 1 1983 - 1992	Passat B1 1978 - 1980
Derby 1 1975 - 1981	Passat B2 1982 - 1988
Derby 2 1982 - 1990	Passat B2 syncro 1985 - 1988
Golf 1 1979 - 1984	Polo 1 1975 - 1981
Golf 1 Cabrio 1979 - 1984	Polo 2 1982 - 1990
Golf 2 1984 - 1989	Rallye Golf 1986 - 1991
Golf 2 syncro 1986 - 1991	Santana 1982 - 1985
Golf Country 1986 - 1991	Scirocco 1 1979 - 1981
Jetta 1 1979 - 1984	Scirocco 2 1981 - 1992
Jetta 2 1984 - 1989	

Not every model year has this gauge!

Technical specifications

supply voltage:

10 V
(LDO)
~1 mA
25 mA
0,5 Hz +/-20%
4,8 Hz +/-20%
-40°C ... 85°C
>83 dBA SPL
(10cm Entfernung)

typ. supply current „buzzer off“:
typ. supply current „buzzer on“:
min. buzzer-frequency:
max. buzzer-frequency:
Operating temperature:
loudness:

Board mounting

tools required:

- 7mm combination wrench
- 7mm socket
- small screwdriver (- or +)



The first step is to remove the instrument cluster !

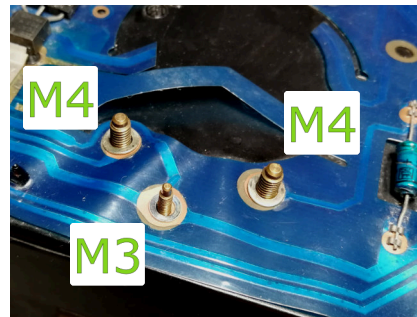


1



loosen the nuts on the rear using a 7mm wrench or 7mm socket (1x M3, 2x M4)

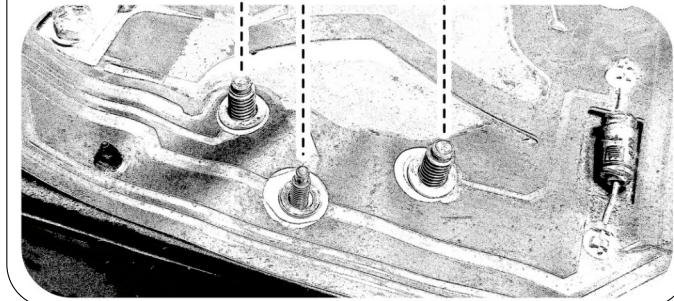
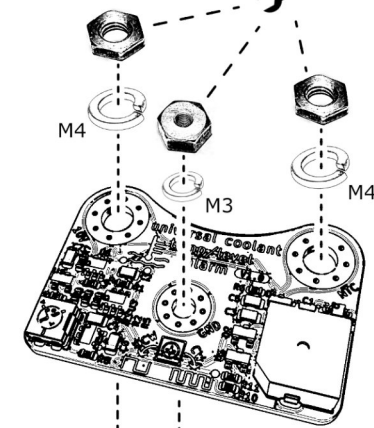
2



check bent washers for presence

3

wrench size 7mm



4



The buzzer-frequency can be adjusted by changing the **potentiometer**-position with a screwdriver (0,5 Hz - 4,8 Hz).



decrease buzzer-frequency



increase buzzer-frequency

universal coolant temp/level alarm V1.0

T3/vanagon 79-92		
Golf-1 Cabrio 79-84		
Golf Country 86-91		
Rallye Golf 86-91		
Santana 82-85		
Scirocco 1 79-81		Golf 1 79-84
Scirocco 2 81-92		Golf 2 84-89
Jetta 1 79-84		Derby 1 75-81
Jetta 2 84-89		Derby 2 82-90
Jetta 2 syncro 86-91		Polo 1 75-81
Passat B1 78-80		Polo 2 82-90
Passat B2 82-88		Passat B2 syncro 85-88
Golf 2 syncro 86-91		
Caddy 1 83-92		

Q.C. _____

Function test

The ignition key must be turned to the first position "Ignition on". The red LED flashes briefly.

Now the respective tests can be performed in the engine bay.



Depending on the car model, the temperature or level sensor is installed at different positions.

The test illustrated here was carried out on a VW T3 CS 1982.

T Overtemperature test

The temperature sensor for the display on the instrument cluster is located in the engine bay on the cooling water flange (the location of the sensor depends on the engine model).

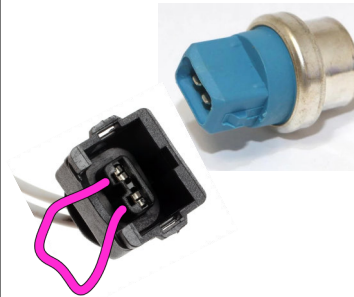
There are several temperature sensor versions.

single-pole water temperature sensor



disconnect the **cable lug** from the temperatur sensor and hold it on **ground** (engine block, gearbox, chassis etc)

two-pole water temperature sensor



disconnect the plug and short-circuit both internal contacts of the plug with a **wire**

test- result:



- temperature needle indicates **maximum** value
- red LED flashes
- buzzer sounds



Be sure to reconnect the plug/cable lug to the temperature sensor after the test has been carried out !

L Coolant level test

In the engine compartment, there is a plug for the level sensor on the coolant reservoir.

After this plug has been disconnected, the electronics signal a loss of water after a few seconds.



disconnect the **plug** of the coolant level sensor and wait a few seconds

test- result:



- temperature needle doesn't chance
- red LED flashes
- buzzer sounds



Be sure to reconnect the plug to the coolant level sensor after the test has been carried out !